

Spectroscopy and Atomic Physics

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Prerequisite Knowledge

- Photons and the idea of quantized energy (see the Electromagnetic Theory & Quantum Physics handout)
- Basic thermal physics: temperature, rms speed of gas particles (see the Thermodynamics handout)
- Basic wave properties: wavelength, frequency, $c = f\lambda$

1 Absorption

Why Atoms Only Accept Specific Photons

An electron bound to an atom can't have just any energy — it's confined to a discrete ladder of allowed energy levels, a direct consequence of quantum mechanics (the same kind of "quantization" we met with Planck's oscillators in the Electromagnetic Theory & Quantum Physics handout, now applied to bound electrons rather than radiation modes). We can derive this ladder explicitly for the simplest atom, hydrogen, using a mix of classical mechanics and one quantum assumption — the approach Bohr used, and one still worth reconstructing by hand.

Model the electron as orbiting the proton in a circular orbit of radius r . The Coulomb force supplies the centripetal force:

$$\frac{ke^2}{r^2} = \frac{m_e v^2}{r} \implies m_e v^2 = \frac{ke^2}{r}.$$

Bohr's key postulate was that angular momentum is quantized in integer multiples of $\hbar$:

$$m_e v r = n\hbar, \quad n = 1, 2, 3, \dots$$

Solving these two equations simultaneously for r (eliminate v using the second equation, substitute into the first):

$$r_n = \frac{n^2 \hbar^2}{k m_e e^2}.$$

The total energy (kinetic plus Coulomb potential) is $E = \frac{1}{2}m_e v^2 - \frac{ke^2}{r} = \frac{ke^2}{2r} - \frac{ke^2}{r} = -\frac{ke^2}{2r}$. Substituting r_n :

$$E_n = -\frac{m_e k^2 e^4}{2\hbar^2} \cdot \frac{1}{n^2} = -\frac{13.6 \text{ eV}}{n^2}.$$

This ladder of allowed energies is the whole reason absorption lines exist. A photon can only be absorbed by an atom in level n if the photon's energy hf exactly matches the gap to some higher level n' : $hf = E_{n'} - E_n$. A photon with slightly too much or too little energy simply isn't absorbed at all — there's no intermediate level for the electron to land on. It's a bit like a vending machine that only accepts exact change in a few specific denominations: close doesn't count, only exact matches go through.

Reading a Star's Ingredients From Its Absorption Lines

Every element has its own unique ladder of energy levels (more complicated than hydrogen's simple formula once there's more than one electron, but still a fixed, discrete, element-specific set of gaps). This means the pattern of absorption lines in a star's spectrum is a direct fingerprint of which elements are present in its outer atmosphere, in exactly the same way a barcode identifies a specific product: see a set of lines matching hydrogen's characteristic gaps, and you know hydrogen is present and absorbing at that layer, regardless of how far away the star is. This is, quite literally, how we know what stars are made of without ever obtaining a physical sample.

2 Emission

The Reverse Process

Emission is absorption run backwards: an electron sitting in an excited level n' drops down to a lower level n , and the energy difference is carried away as a photon of frequency $f = (E_{n'} - E_n)/h$. This can happen **spontaneously** (the electron just decays after some characteristic lifetime, with no external trigger) or via **stimulated emission** (a passing photon of the right energy triggers the transition early, producing a second photon identical in energy, direction, and phase to the first — the physical basis of lasers, though not something we'll need further here).

The Rydberg Formula Using the Bohr energy levels derived above, the photon energy released dropping from n' to n (with $n' > n$) is

$$hf = E_{n'} - E_n = 13.6 \text{ eV} \left(\frac{1}{n^2} - \frac{1}{n'^2} \right).$$

Converting to wavelength via $f = c/\lambda$ and rearranging:

$$\boxed{\frac{1}{\lambda} = R \left(\frac{1}{n^2} - \frac{1}{n'^2} \right)}, \quad R \equiv \frac{13.6 \text{ eV}}{hc} \approx 1.097 \times 10^7 \text{ m}^{-1}$$

the **Rydberg formula**, matching (to excellent precision) the observed hydrogen spectral series: $n = 2$ gives the Balmer series (visible light, the famous $H\alpha$, $H\beta$, ... lines), $n = 1$ gives the Lyman series (ultraviolet), and $n = 3$ gives the Paschen series (infrared).

Nebulae as Emission-Line Objects

A hot young star can ionize the hydrogen gas surrounding it, stripping electrons entirely from their atoms. As free electrons recombine with protons, they cascade back down through the energy ladder, level by level, emitting a photon at each step — and since this happens throughout an enormous, sparse cloud of gas with no dense stellar surface behind it to provide a background continuum, the result is a spectrum made almost *entirely* of bright emission lines (most famously $H\alpha$, which is why star-forming regions glow characteristically red-pink in long-exposure images). This is exactly what an emission nebula is, physically: a vast recombination cascade, lit by a nearby hot star, glowing at very specific, predictable wavelengths.

3 Scattering

Scattering Is Not Absorption

In absorption, a photon's energy is entirely taken up by the atom and (eventually, via a potentially different re-emission process) may reappear as a completely different photon. In **scattering**, the photon interacts with a charge and re-emerges essentially immediately, typically in a new direction, without ever being "digested" into an internal atomic transition.

Why the Sky is Blue: Rayleigh Scattering

When light passes an electron bound tightly in an atom (much smaller than the light's wavelength), the oscillating electric field of the light wave shakes the electron back and forth, turning it into a

tiny oscillating dipole antenna, re-radiating at the *same* frequency it was driven at. An oscillating dipole radiates power proportional to the square of its acceleration, and since the electron is driven at frequency f , its acceleration scales as f^2 , so the radiated power scales as f^4 — or equivalently, since $f \propto 1/\lambda$,

$$P_{\text{scattered}} \propto \frac{1}{\lambda^4}.$$

This is **Rayleigh scattering**, and that steep $1/\lambda^4$ dependence is dramatic: blue light ($\lambda \approx 450$ nm) scatters roughly $(700/450)^4 \approx 5.8$ times more strongly than red light ($\lambda \approx 700$ nm) off the same air molecules. Sunlight scattered many times across the whole sky is therefore disproportionately blue — which is why the sky is blue, and, by the same logic in reverse, why sunsets are red (the direct sunlight reaching your eyes at sunset has had its blue component preferentially scattered *away* along the long, low path through the atmosphere, leaving mostly red light to get through directly).

Thomson Scattering: Off Free Electrons

Free electrons (not bound to any atom, common in the hot, ionized outer layers of stars) also scatter light, but since there's no restoring force pulling them back to an equilibrium position, the physics is slightly different, and the scattering cross-section, $\sigma_T \approx 6.65 \times 10^{-29} \text{ m}^2$, comes out *independent* of wavelength.

Closing the Loop on the Thermodynamics Handout

This is exactly the kind of scattering responsible for the mean free path we used, without deriving it, in the random-walk photon diffusion estimate in the Thermodynamics handout. In the hot, fully ionized layers of a star, electron (Thomson) scattering is often the dominant source of opacity, setting how far a photon can travel before its direction is randomized — the very quantity ℓ that determined how many thousands of years it takes sunlight to random-walk its way out of the Sun.

4 Spectra of Celestial Objects

Kirchhoff's Three Laws

Put absorption, emission, and scattering together, and you get a complete classification of every spectrum you'll ever see, first worked out empirically by Gustav Kirchhoff:

1. A hot, dense object (solid, liquid, or dense gas) emits a **continuous spectrum** — light at all wavelengths, with no gaps or spikes. At high density, atoms are so close together and colliding so frequently that their energy levels smear into an effective continuum, and the material behaves essentially like the blackbody radiator from the Electromagnetic Theory & Quantum Physics handout.
2. A hot, low-density gas produces an **emission-line spectrum** — bright lines at specific wavelengths, and nothing in between. With no dense background continuum to compete against, and atoms far enough apart to keep their sharp, discrete energy levels, you only see light at the exact wavelengths corresponding to their allowed transitions.

3. A continuous spectrum, viewed *through* a cooler, low-density gas, produces an **absorption-line spectrum** — a continuum with narrow dark gaps at exactly the wavelengths the intervening gas is capable of absorbing.

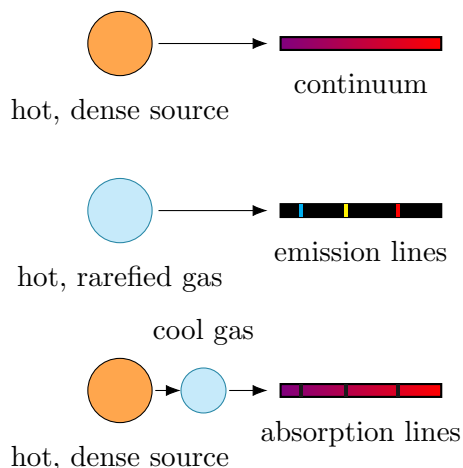


Figure 1: Kirchhoff's three laws. Top: a hot dense source alone gives a continuum. Middle: a hot rarefied gas alone gives emission lines. Bottom: a continuum source viewed through an intervening cooler gas gives an absorption-line spectrum — exactly the situation for an ordinary star.

This Is Literally What a Star's Spectrum Is

A typical star's spectrum is Kirchhoff's third case, playing out in a single object: the dense, hot photosphere below produces a near-continuous (near-blackbody) spectrum, and the cooler, less dense layers of gas just above it (still part of the same star) selectively absorb specific wavelengths on the way out, imprinting the dark absorption lines we actually observe and use to determine composition (Section 1). Meanwhile, the Sun's much hotter but far less dense corona, sitting even farther out, produces its own faint emission-line spectrum (Kirchhoff's second case) — which is why, historically, some coronal emission lines were initially so unfamiliar that they were attributed to a hypothetical new element, "coronium," before physicists realized they were just highly ionized iron, doing something no laboratory on Earth could yet replicate.

5 The Doppler Effect

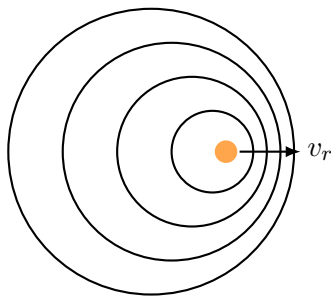
Deriving the Shift From Scratch

Suppose a source emits light of true wavelength λ_0 (period $T_0 = \lambda_0/c$ in its own rest frame), while moving directly away from an observer at radial speed v_r . Consider two successive wave crests. The source emits the first crest, then waits one period T_0 , and emits the second. But during that time, the source has moved a further distance $v_r T_0$ away. The second crest therefore has to travel this extra distance to catch up to where the first crest would be, stretching the effective wavelength the observer measures:

$$\lambda_{\text{obs}} = cT_0 + v_r T_0 = \lambda_0 \left(1 + \frac{v_r}{c}\right) \implies \boxed{\frac{\Delta\lambda}{\lambda_0} = \frac{v_r}{c}} \quad (v_r \ll c),$$

with $\Delta\lambda = \lambda_{\text{obs}} - \lambda_0$ positive (redshifted, longer wavelength) for a source moving away, and negative (blueshifted) for a source moving toward the observer. Only the *radial* component of velocity

matters — motion purely across the line of sight produces no first-order Doppler shift at all, since it doesn't change the light-travel-time argument above.



crests bunch up ahead,
spread out behind

Figure 2: A moving source's wavefronts compress on the side it's moving toward (blueshift) and stretch on the side it's moving away from (redshift) — exactly like the changing pitch of a passing ambulance siren, but for light instead of sound.

Nearly Everything in Dynamical Astronomy

The Doppler effect is arguably the single most exploited tool in observational astrophysics. Recall from the Celestial Mechanics handout: a binary star's individual orbital speeds v_1, v_2 — which directly give the mass ratio M_2/M_1 — are measured by tracking exactly this kind of periodic redshift-then-blueshift oscillation in each star's spectral lines as they orbit their common barycentre. Exoplanets are routinely detected the same way, by catching the tiny reflex wobble a planet induces in its host star. On a cosmological scale, the redshift of distant galaxies' spectral lines is the observational basis of Hubble's law (though, importantly, the *cosmological* redshift of far-away galaxies is technically due to the expansion of space itself stretching light in transit, not a simple Doppler shift from motion *through* space — a subtlety worth keeping straight, even though the two produce a mathematically similar-looking $\Delta\lambda/\lambda$ at low redshift). And within a single star, different points across its visible disk — one edge rotating toward us, the other away — produce a whole range of Doppler shifts simultaneously, which is precisely what causes rotational line broadening, covered fully in Section 8.

6 Line Formation

Selection Rules and “Forbidden” Lines

Not every conceivable jump between energy levels is actually allowed to happen via the ordinary, fast process of spontaneous photon emission — quantum mechanical **selection rules** (governing how the electron's angular momentum can change during a transition) forbid some jumps outright, or at least make them extraordinarily improbable via direct radiative decay. An electron stuck in one of these **metastable** states can survive for a comparatively enormous time (seconds, or even longer, compared to the nanosecond timescales of ordinary allowed transitions) before it manages to decay anyway, emitting what's called a **forbidden line**.

Why Nebulae Show Lines You'll Never See in a Lab

Under everyday laboratory densities, an atom sitting in a metastable state almost never gets the chance to emit its forbidden line — it gets knocked out of that state by a collision with another particle long before the slow radiative decay has a chance to occur. But in the extraordinarily low-density gas of a planetary nebula or supernova remnant (far emptier than any laboratory vacuum), collisions become so rare that an atom might wait undisturbed for the necessary seconds (or longer) for the forbidden transition to finally happen. This is exactly why famous nebular lines like [O III] (doubly-ionized oxygen) and [N II] show up prominently in nebular spectra (conventionally written with square brackets to flag them as forbidden) but are essentially never seen from dense laboratory sources — they're not exotic physics, just ordinary transitions that need an emptiness no lab can practically provide, but which interstellar space offers routinely.

7 Continuum Spectra

Beyond the Blackbody: Free Electrons Make Continuum Light Too

A star's overall continuum comes mostly from the near-blackbody radiation of its dense photosphere (Electromagnetic Theory & Quantum Physics handout). But hot, ionized gas — like a nebula, or a stellar corona — has no dense photosphere at all, and yet it still produces a genuine, if fainter, continuum, through two distinct free-electron processes:

Free-free emission (bremsstrahlung, German for “braking radiation”): a free electron passing near an ion is deflected and decelerated by the ion's electric field, and any decelerating charge radiates. Since the electron can be deflected by any amount, losing any amount of energy, the emitted photons can have essentially any energy at all — a genuine continuum, with no preferred wavelength.

Free-bound emission (recombination continuum): a free electron with kinetic energy KE is captured directly into a bound level E_n of an ion, and the photon carries away both the binding energy *and* the leftover kinetic energy: $hf = |E_n| + KE$. Since KE can take any value, this also produces a continuum — but with a sharp edge (the **recombination edge**, or in hydrogen's case, most famously the **Balmer jump** at the series limit for recombination onto $n = 2$), since photons below the pure binding-energy threshold simply can't be produced by capture into that particular level.

The Balmer Jump as a Temperature and Gravity Diagnostic

The size of the Balmer jump — the abrupt discontinuity in a star's continuum right at the Balmer series limit — depends sensitively on the temperature and density of the star's photosphere (since it governs the balance of hydrogen atoms sitting in the $n = 2$ level, available to photoionize, versus already-ionized free electrons available to recombine into it). Measuring the size of this jump is a genuinely useful, independent way to estimate a star's temperature and surface gravity, entirely separate from fitting the overall blackbody-like continuum shape or measuring individual absorption line strengths.

8 Splitting and Broadening of Spectral Lines

Zeeman Splitting: Lines as Magnetometers

A Classical Estimate of Zeeman Splitting

An orbiting electron is a tiny current loop, and current loops have a magnetic moment. Placed in an external magnetic field B , this magnetic moment experiences a torque, causing the electron's orbit to **precess** around the field direction — exactly like a spinning top precessing under gravity — at the **Larmor frequency**. Setting the magnetic torque equal to the rate of change of angular momentum for this precession (a standard gyroscopic argument) gives

$$\omega_L = \frac{eB}{2m_e}.$$

This precession adds or subtracts a small extra frequency shift $\pm\omega_L/2\pi$ to the electron's effective orbital motion, splitting what was a single spectral line into multiple closely-spaced components separated by

$$\Delta f \approx \frac{eB}{4\pi m_e}.$$

(The full quantum treatment refines the numerical factors and predicts more than just three components in general, but this classical estimate captures the correct scaling: splitting proportional to B , and inversely proportional to the electron's mass.)

Weighing a Magnetic Field From Millions of Kilometres Away

Since the splitting is directly proportional to B , measuring how far a spectral line splits (or, more practically, how much it broadens or shows distinct polarized components when a full splitting can't be cleanly resolved) gives a direct measurement of the magnetic field strength at the source — no other technique lets you weigh a magnetic field from light-years away. This is exactly how sunspot magnetic fields were first measured (Hale, 1908, finding sunspots to be sites of intensely concentrated magnetic field, which is also *why* they're cool and dark: strong magnetic fields there suppress the convective energy transport discussed in the Thermodynamics handout), and it remains a standard tool for measuring magnetic fields on other stars entirely.

Broadening Mechanism 1: Natural (Lifetime) Broadening

An excited state doesn't live forever — it has some finite mean lifetime τ before it decays. The energy-time uncertainty principle, $\Delta E \Delta t \gtrsim \hbar$, then tells us that a state with a finite lifetime τ cannot have a perfectly sharp energy — there's an inherent uncertainty $\Delta E \sim \hbar/\tau$, which translates into an inherent minimum linewidth

$$\Delta f \sim \frac{1}{2\pi\tau}$$

that no amount of instrumental precision could ever remove — it's not a measurement limitation, but a fundamental property of the transition itself.

Broadening Mechanism 2: Doppler (Thermal) Broadening

Not every atom in a gas is at rest — they have a whole spread of random thermal velocities, with rms speed $v_{\text{rms}} = \sqrt{3kT/m}$ (derived from scratch in the Thermodynamics handout). Each atom's

individual, randomly-directed velocity gives its absorbed or emitted light its own small individual Doppler shift (Section 5), and since the velocities are randomly distributed in every direction, the line doesn't just shift — it *smears out* into a range of wavelengths, with a characteristic width

$$\frac{\Delta\lambda}{\lambda} \sim \frac{v_{\text{rms}}}{c} = \frac{1}{c} \sqrt{\frac{3kT}{m}}.$$

Because the underlying distribution of thermal velocities is the familiar bell-curve Maxwell-Boltzmann distribution, the resulting line profile is a **Gaussian**.

Broadening Mechanism 3: Pressure (Collisional) Broadening

In a denser gas, atoms collide with each other frequently, and each collision can abruptly disrupt an ongoing emission or absorption process, effectively shortening the coherence time available for a clean transition — much like the natural-lifetime argument above, but now the relevant timescale is the time between collisions rather than the intrinsic quantum lifetime. This produces a line profile with a different characteristic shape (a **Lorentzian**, with more extended wings than a Gaussian), and its width grows directly with the surrounding gas density and pressure.

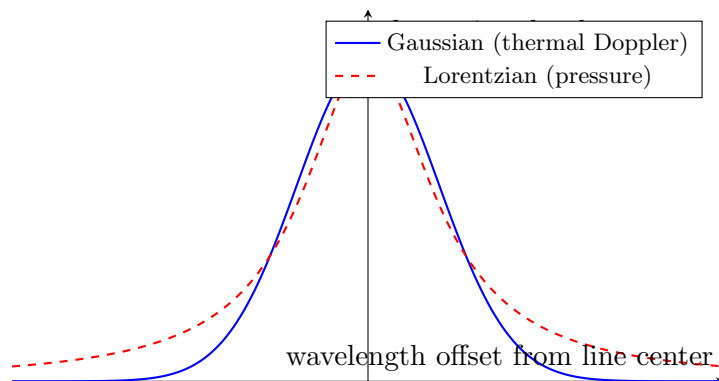


Figure 3: Thermal (Doppler) broadening gives a Gaussian profile, sharply falling off away from line center; pressure broadening gives a Lorentzian, with a similar core but much more extended wings — the two are distinguishable in real spectra precisely because of this difference in wing shape.

Telling Giants From Dwarfs at the Same Temperature

Two stars can have identical surface temperatures (identical Wien's-law color, identical overall line pattern) and yet be wildly different objects — a giant and a main-sequence dwarf — and their spectral lines can still tell them apart. A giant star has a much larger radius and hence a far lower photospheric density and pressure than a dwarf of the same temperature (recall from the Celestial Mechanics handout that $L \propto R^2 T^4$: for the same T , a much larger L implies a much larger R , spreading the same mass over a far larger volume). Lower pressure means weaker pressure broadening, so a giant's absorption lines are noticeably narrower and sharper than a dwarf's at the same temperature — a genuine, measurable **luminosity/gravity diagnostic** hiding entirely in line *shape*, independent of anything about line *position* (temperature/composition) at all. Meanwhile, a star's overall rotation adds yet another, independent broadening contribution: different points across the visible stellar disk move toward or away from us at different line-of-sight speeds (Section 5), broadening every line by an amount that grows with the star's rotational velocity — letting us measure how fast a star spins without ever resolving its disk.

9 Polarisation

What Polarisation Means, and Why Scattering Produces It

Ordinary starlight is **unpolarised**: the electric field oscillates in essentially random, constantly-changing directions perpendicular to the direction of travel. **Polarised** light has its electric field oscillating in a preferred, consistent orientation (linear polarisation) or rotating in a systematic pattern (circular/elliptical polarisation).

Scattering is a natural way to *produce* polarisation from unpolarised light, and the reason is geometric. Recall from Section 3 that scattering works by an incident electric field driving an electron into oscillation, which then radiates as an oscillating dipole. A crucial property of dipole radiation is that it radiates *no* power directly along the axis it's oscillating along — all the radiated power goes out perpendicular to that axis. So if you observe scattered light at exactly 90° to the original, incoming direction, you can only be receiving light from the components of the electron's oscillation that are perpendicular to *your* line of sight — and it turns out only one single oscillation direction satisfies "perpendicular to both the original beam and your line of sight" simultaneously. The scattered light you see at 90° is therefore **completely, perfectly linearly polarised**, even though the original light was completely unpolarised.

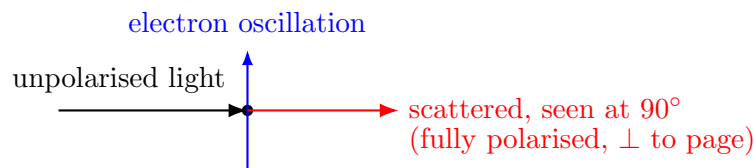


Figure 4: An electron driven by unpolarised light oscillates along the field direction (blue) and radiates like a dipole. Viewed at 90° to the original beam (red), only one oscillation direction can contribute — the scattered light is fully polarised.

Stellar and Astrophysical Applications

Polarisation shows up as a genuinely independent diagnostic across a huge range of astronomical contexts:

- **Circumstellar and interstellar dust:** starlight scattered off dust grains in circumstellar disks or the interstellar medium becomes partially polarised by exactly the mechanism above, letting astronomers map out otherwise invisible disk and dust geometry.
- **Interstellar magnetic fields:** elongated dust grains tend to align their long axes perpendicular to the local interstellar magnetic field (their rotation is damped preferentially along that axis). Starlight passing through then gets slightly polarised in a direction tracing the field — turning polarimetry into one of the only direct ways to map magnetic field *direction* across the galaxy.
- **The Zeeman effect, revisited:** the different split components from Section 8 aren't just shifted in wavelength — they emerge with different, specific polarisation states, which is in fact how solar and stellar magnetic fields are measured in practice (spectropolarimetry), not merely by linewidth alone.
- **Synchrotron radiation:** relativistic electrons spiraling around magnetic field lines (common in jets from active galactic nuclei, supernova remnants, and pulsar wind nebulae) emit

strongly polarised radiation, with the polarisation direction directly tracing the magnetic field geometry in these otherwise inaccessible, extreme environments.

10 Summary Sheet

Topic	Key Result
Bohr energy levels	$E_n = -13.6 \text{ eV}/n^2$
Rydberg formula	$1/\lambda = R(1/n^2 - 1/n'^2)$
Rayleigh scattering	$P \propto 1/\lambda^4$ (why the sky is blue)
Kirchhoff's laws	dense $\rightarrow$ continuum; rarefied hot $\rightarrow$ emission; continuum + cool gas $\rightarrow$ absorption
Doppler shift	$\Delta\lambda/\lambda_0 = v_r/c$
Forbidden lines	metastable states, only visible at very low density (nebulae)
Continuum beyond blackbody	free-free (bremsstrahlung) and free-bound (recombination) emission
Zeeman splitting	$\Delta f \approx eB/(4\pi m_e)$ — measures magnetic field strength
Natural broadening	$\Delta f \sim 1/(2\pi\tau)$, from $\Delta E\Delta t \gtrsim \hbar$
Thermal (Doppler) broadening	$\Delta\lambda/\lambda \sim v_{\text{rms}}/c$; Gaussian profile
Pressure broadening	grows with density; Lorentzian profile; distinguishes giants from dwarfs
Polarisation from scattering	light scattered at 90° is fully linearly polarised